WORKED SOLUTION

STEP II 2003, Question 6

Pure • Integration

STEP II 2003, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q6

The key idea

Applying f once more means "subtract 1, then reflect anything negative back above the axis". Do that to a graph whose values on some stretch lie between 0 and 1, and that stretch is simply turned upside down inside the strip $0 \le y \le 1$, while the long straight arms gain one extra tooth. So the graphs become zigzags (triangle waves), and once you see that, both integrals in the final part are just sums over identical periods.

Solution

The sketches

Two facts do all the work.

Recurrence. Since $g_{n+1} = f \circ g_n$, we have $g_{n+1}(x) = |g_n(x) - 1|$: shift the graph of g_n down by 1 and reflect the parts below the x-axis.

Symmetry. Since $f(2-x) = |1-x| = f(x)$, we get $g_n(2-x) = g_{n-1}(f(2-x)) = g_{n-1}(f(x)) = g_n(x)$ for $n \ge 2$, and directly for $n = 1$. So every g_n is symmetric about the line $x = 1$, and it is enough to understand $x \ge 1$.

Start with $g_1(x) = |x-1|$, a V with vertex $(1, 0)$.

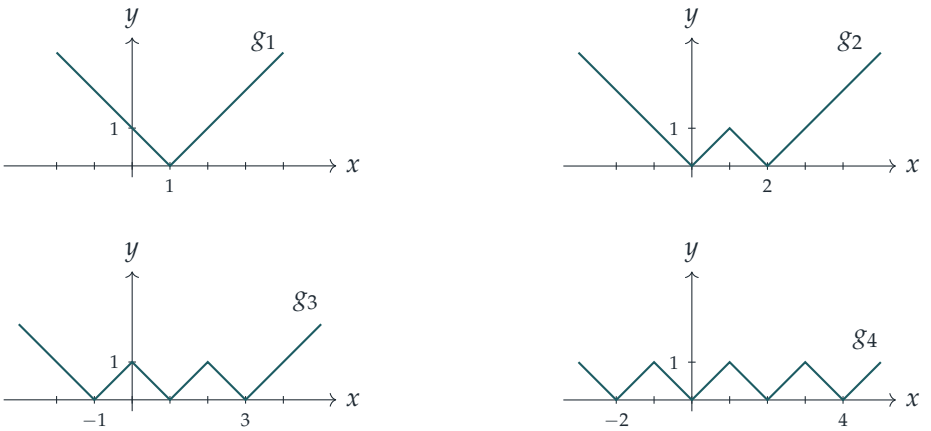
For g_2: on $[0, 2]$ we have $0 \le g_1 \le 1$, so $g_2 = 1 - |x-1|$, a tent rising to 1 at $x = 1$ and falling to 0 at $x = 0$ and $x = 2$. For $x \ge 2$, $g_2 = x - 2$; by symmetry $g_2 = -x$ for $x \le 0$. So g_2 is a W shape.

Each further step does the same thing: the zigzag part (with values in $[0, 1]$) is turned upside down, and the straight arm $x - n$ becomes $|x - n - 1|$, which adds one more tooth on $[n, n+1]$ (and, by symmetry, on $[1-n, 2-n]$). Writing $d_{\text{even}}(x)$ and $d_{\text{odd}}(x)$

for the distance from x to the nearest even and nearest odd integer, this gives

$$g_n(x) = \begin{cases} 2-n-x, & x \le 2-n, \\ d_{\text{even}}(x) \text{ if } n \text{ is even, } d_{\text{odd}}(x) \text{ if } n \text{ is odd}, & 2-n \le x \le n, \\ x-n, & x \ge n. \end{cases}$$

This follows by induction from the recurrence: $d_{\text{even}} + d_{\text{odd}} = 1$ for every real x (between consecutive integers the two distances are t and $1-t$), so "1 minus the zigzag" swaps the parity; and on $[n, n+1]$ the new piece $n+1-x$ is exactly the distance to $n+1$, the nearest integer of the new parity. The left-hand side follows by symmetry about $x = 1$.



In words: g_3 has zeros at $-1, 1, 3$ and peaks of height 1 at 0 and 2, with arms $y = -1 - x$ and $y = x - 3$; g_4 has zeros at $-2, 0, 2, 4$ and peaks of height 1 at $-1, 1, 3$, with arms $y = -2 - x$ and $y = x - 4$. All four graphs are symmetric about $x = 1$.

The integral

Let n be even. Since $n \ge 2$, the interval $[0, n]$ lies inside $[2-n, n]$, so on it $g_n(x) = d_{\text{even}}(x)$.

Ask yourself Why is it enough to integrate over $[0, 2]$?

Both functions have period 2: $d_{\text{even}}(x+2) = d_{\text{even}}(x)$ because shifting by 2 maps even integers to even integers, and $\left|\sin \frac{\pi(x+2)}{2}\right| = \left|-\sin \frac{\pi x}{2}\right| = h(x)$. As n is even, $[0, n]$ is exactly $\frac{n}{2}$ whole periods $[0,2], [2,4], \ldots, [n-2, n]$, so each integral over $[0, n]$ is $\frac{n}{2}$ times the integral over $[0, 2]$.

On $[0, 2]$, g_n is the tent $1 - |x-1|$: a triangle with base 2 and height 1, so

$$\int_0^2 g_n(x)\,\mathrm{d}x = \tfrac{1}{2} \cdot 2 \cdot 1 = 1.$$

On $[0,2]$, $\frac{\pi x}{2} \in [0,\pi]$, so $\sin\frac{\pi x}{2} \geq 0$ and the modulus can be dropped:

$$\int_0^2 h(x)\,\mathrm{d}x = \int_0^2 \sin\frac{\pi x}{2}\,\mathrm{d}x = \Big[-\frac{2}{\pi}\cos\frac{\pi x}{2}\Big]_0^2 = \frac{2}{\pi}+\frac{2}{\pi} = \frac{4}{\pi}.$$

Therefore

$$\int_0^n \big(h(x)-g_n(x)\big)\,\mathrm{d}x = \frac{n}{2}\Big(\frac{4}{\pi}-1\Big) = \frac{2n}{\pi}-\frac{n}{2},$$

as required.

Checks

- Pick a value and iterate by hand: $g_4(-1)$: $-1 \to 2 \to 1 \to 0 \to 1$, so $g_4(-1) = 1$, a peak, as in the sketch.
- The answer is positive, since $4/\pi \approx 1.27 > 1$: the sine arch bulges above the straight-sided tent with the same base and height, which matches the picture of the two graphs.
- Numerically, Simpson's rule for $n = 2$ gives 0.273240, agreeing with $\frac{4}{\pi} - 1 \approx 0.273240$; the same agreement holds for $n = 4, 6, 8, 10$.

Common slips

- Forgetting the left-hand side: for $x < 0$ the graphs are not straight lines once $n \geq 3$. The symmetry $g_n(2-x) = g_n(x)$ is the quickest way to get them right.
- Integrating $\sin\frac{\pi x}{2}$ over $[0,n]$ without the modulus, which gives 0 for even n instead of $\frac{2n}{\pi}$.

Beyond STEP

Iterating a simple map and watching what the graphs do is the starting point of discrete dynamical systems. Here the zigzags settle into a triangle wave, and the closely related "tent map" is a classic example of chaos. The comparison of h with the triangle wave also points to Fourier series, where a triangle wave is written as a sum of sines and cosines, and $|\sin|$ is one of the first examples. Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction to Fourier series.

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